8 The International Space Station (ISS) is a habitable artificial satellite. It is maintained at 340 km above the Earth's surface, in what is known as a Low Earth Orbit (LEO). The radius of the Earth is 6.37×10^6 m. Fig 8.1 shows how the gravitational field strength of the Earth, *g*, varies with distance from the Earth's surface.

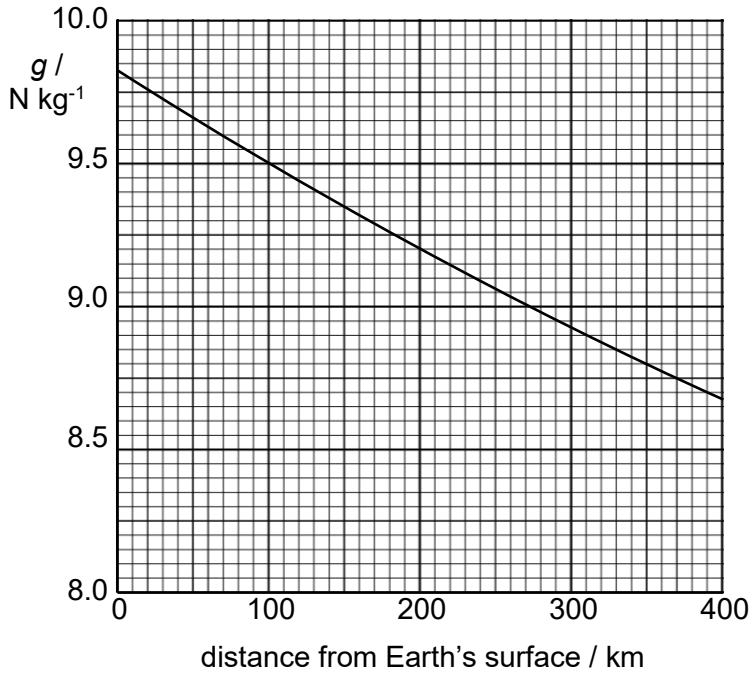

Fig. 8.1

(a) Using Fig 8.1, explain why an astronaut experiences apparent weightlessness in the ISS despite a non-zero gravitational field strength at LEO.

.....

.....

.....

..... [2]

(b) Astronauts in apparent weightlessness may lose muscle mass. Fig. 8.2 shows a model of a spring system which can be used to monitor changes in mass.

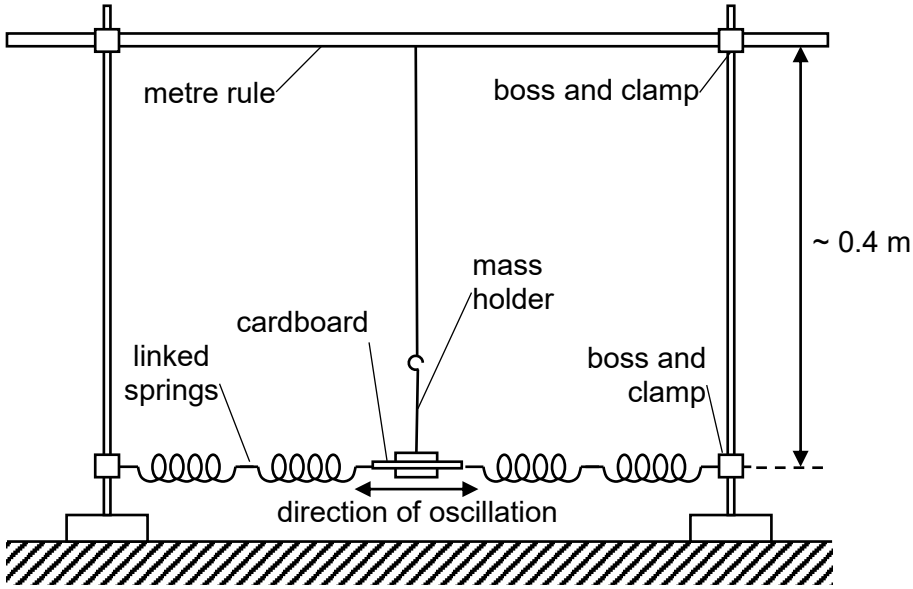

Fig. 8.2

It has been suggested that the relationship between the period of oscillation T and the mass M is given by:

$$\frac{1}{T^2} = \frac{p}{M} + q$$

where p and q are constants.

Fig. 8.3 shows the experimental results obtained using a stopwatch to measure the time taken for oscillations.

M / kg	time taken for 20 oscillations		T / s	$\frac{1}{T^2}$ / s^{-2}	$\frac{1}{M}$ / kg^{-1}
	t_1 / s	t_2 / s			
0.100	10.34	10.38	0.5180	3.727	10.0
0.150	12.07	11.93	0.6000	2.778	6.67
0.200	13.39	13.43	0.6705	2.242	5.00
0.250	14.34	14.32	0.7165	1.948	4.00
0.300	15.23	15.23	0.7615	1.724	3.33
0.350	15.70	15.64	0.7835	1.629	2.86

Fig. 8.3

The corresponding values of $\frac{1}{T^2}$ and $\frac{1}{M}$ are plotted on the graph in Fig. 8.4.

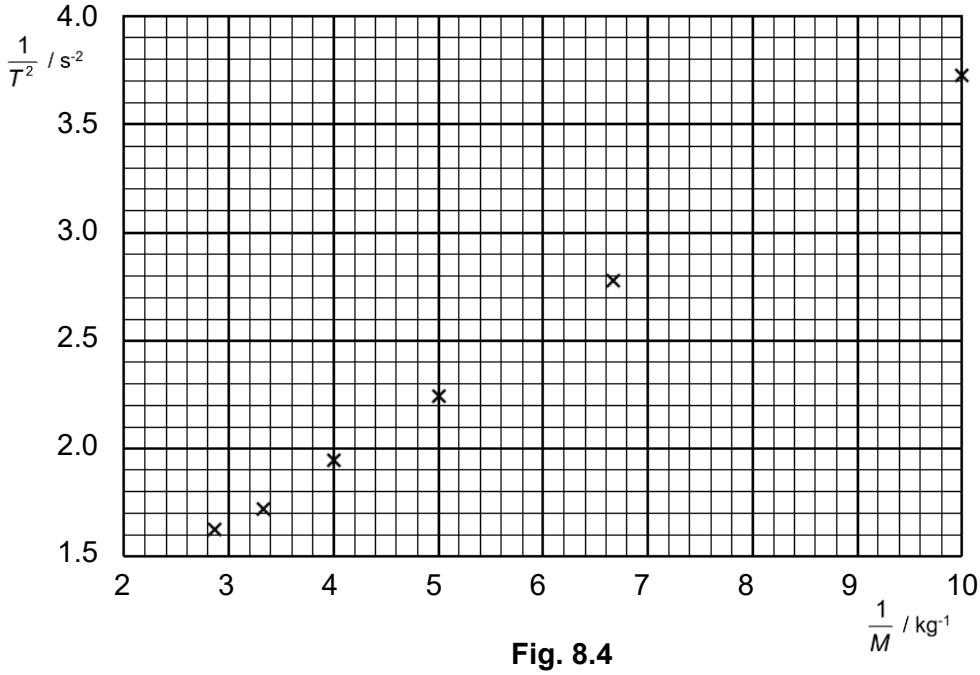

Fig. 8.4

(i) Using Fig. 8.4, determine *p* and *q*.

p = ...

q = ... [3]

(ii) It has been suggested that the variation of period, ∂T, is related to an astronaut's mass loss by $\partial T = \frac{2pT^3}{M} \times (\text{fractional mass loss})$.

Determine ∂T corresponding to an initial mass of M = 0.5 kg and a fractional mass loss of 0.10.

∂T = ...s [2]

(iii) According to medical opinion, a muscular mass loss of 10% is concerning. Taking reference from the data available, discuss if the experiment is able to detect the variation ∂T for a mass as small as 0.5 kg.

.....

.....

..... [2]